Extra Bite #12: Electro-Optical Devices

Photovoltaic cells, photoconductors, optoisolators, optical shaft encoders, solid-state relays.

Group E6F · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Light Into Electricity (And Back)

Two related effects sit at the heart of this group:

Effect	What happens	Q
Photovoltaic	Light hits a semiconductor junction; photons knock electrons across the band gap; a voltage appears at the terminals. **The photovoltaic effect is the conversion of light to electrical energy.** (E6F04)	E6F04
Photoconductive	Light shines on a semiconductor; freed electrons make the material a better conductor. **Resistance decreases** when light hits a photoconductive material. (E6F02)	E6F02

What absorbs the energy from light falling on a photovoltaic cell? Electrons. (E6F01) Photons surrender their energy to electrons in the semiconductor, kicking them across the band gap into the conduction band. Once there, the built-in junction field sweeps them across, producing voltage at the terminals.

Solar Cell Specifics

Question	Answer	Q
Most common material for power-generating PV cells	**Silicon** (E6F10)	E6F10
Approximate open-circuit voltage of a fully illuminated silicon PV cell	**0.5 V** (E6F11). Why series strings of cells are needed for usable voltages.	E6F11
Efficiency of a PV cell	**The relative fraction of light that is converted to current.** (E6F09)	E6F09

Photoconductive Devices

What material is most commonly used for photoconductive devices? Crystalline semiconductor (E6F06). CdS, Si, Ge – all crystalline semiconductors whose resistance drops when illuminated.

Optical Sensing And Switching

Device	What it does	Q
Optical shaft encoder	**A device that detects rotation by interrupting a light source with a patterned wheel.** (E6F05) The pulse pattern tells the electronics how fast and which direction the shaft is turning.	E6F05
Optoisolator / optocoupler	**An LED on the input side and a phototransistor on the output side**, sealed in one package. (E6F03) Signal crosses the gap as light; no electrical connection between input and output.	E6F03
Solid-state relay	**A device that uses semiconductors to implement the functions of an electromechanical relay.** (E6F07) Often a TRIAC or MOSFET driven through an optoisolator.	E6F07

Why Pair Optoisolators With AC-Line Solid-State Circuits?

Optoisolators provide electrical isolation between a control circuit and the circuit being switched. (E6F08) The 120 VAC side stays galvanically separate from the low-voltage logic side: a fault on the high-voltage side cannot conduct back to your microcontroller. Light is the only thing crossing the isolation gap.

Mental Cheat Sheet

- Photovoltaic effect = light → electrical energy.
- In a PV cell, **electrons** absorb the photon energy.
- Photoconductive material: light → **resistance decreases**.
- Photoconductive material = **crystalline semiconductor**.
- Solar cells are silicon; **open-circuit V ≈ 0.5 V** per cell.
- PV efficiency = fraction of light converted to current.
- Optical shaft encoder = pulsed-light-through-patterned-wheel rotation sensor.
- Optoisolator = LED + phototransistor in one package.
- Solid-state relay = semiconductor doing a relay's job.
- Optoisolator on AC lines: **electrical isolation** between control and load.

Common Traps

- Photons hit **electrons**, not protons or holes. "Protons" is the classic trap answer.
- Photoconductive resistance **decreases** in light. The trap is "increases."
- PV cell silicon open-circuit V is **0.5 V**, not 0.7 V or 1.1 V.
- Optoisolator = LED + phototransistor (not a laser, not a photomultiplier).
- The reason for an optoisolator on AC mains is **electrical isolation**, not impedance matching.
- A solid-state relay does not "use transistors to drive the relay coil" – there is no coil; semiconductors do the switching.

Takeaway: Light interacts with semiconductors two ways: photovoltaic (light → voltage, used in solar cells) and photoconductive (light → lower resistance). Optical shaft encoders detect rotation by chopping a light beam with a patterned wheel; optoisolators are an LED + phototransistor pair that ferry signals across an electrical-isolation gap; solid-state relays do the relay job without any moving parts.

Bite 12 Question Bank (E6F)

E6F01. What absorbs the energy from light falling on a photovoltaic cell?
A. Protons.
B. Photons.
C. Electrons.
D. Holes.

E6F02. What happens to photoconductive material when light shines on it?
A. Resistance decreases.
B. Resistance increases.
C. Reflectivity increases.
D. Reflectivity decreases.

E6F03. What is the most common configuration of an optoisolator or optocoupler?
A. A lens and a photomultiplier.
B. A frequency-modulated helium-neon laser.
C. An amplitude-modulated helium-neon laser.
D. An LED and a phototransistor.

E6F04. What is the photovoltaic effect?
A. The conversion of voltage to current when exposed to light.
B. The conversion of light to electrical energy.
C. The effect that causes a photodiode to emit light when a voltage is applied.
D. The effect that causes a phototransistor's beta to decrease when exposed to light.

E6F05. Which of the following describes an optical shaft encoder?
A. A device that detects rotation by interrupting a light source with a patterned wheel.
B. A device that measures the strength of a beam of light using analog-to-digital conversion.
C. An optical computing device in which light is coupled between devices by fiber optics.
D. A device for generating RTTY signals by means of a rotating light source.

E6F06. Which of these materials is most commonly used to create photoconductive devices?
A. Polyphenol acetate.
B. Argon.
C. Crystalline semiconductor.
D. All these choices are correct.

E6F07. What is a solid-state relay?
A. A relay that uses transistors to drive the relay coil.
B. A device that uses semiconductors to implement the functions of an electromechanical relay.
C. A mechanical relay that latches in the on or off state each time it is pulsed.
D. A semiconductor switch that uses a monostable multivibrator circuit.

E6F08. Why are optoisolators often used in conjunction with solid-state circuits that control 120 VAC circuits?
A. Optoisolators provide a low-impedance link between a control circuit and a power circuit.
B. Optoisolators provide impedance matching between the control circuit and power circuit.
C. Optoisolators provide an electrical isolation between a control circuit and the circuit being switched.
D. Optoisolators eliminate the effects of reflected light in the control circuit.

E6F09. What is the efficiency of a photovoltaic cell?
A. The output RF power divided by the input DC power.
B. The output in lumens divided by the input power in watts.
C. The open-circuit voltage divided by the short-circuit current under full illumination.
D. The relative fraction of light that is converted to current.

E6F10. What is the most common material used in power-generating photovoltaic cells?
A. Selenium.
B. Silicon.
C. Cadmium sulfide.
D. Indium arsenide.

E6F11. What is the approximate open-circuit voltage produced by a fully illuminated silicon photovoltaic cell?
A. 0.5 volts.
B. 0.7 volts.
C. 1.1 volts.
D. 1.5 volts.

Answer Key

E6F01: C	E6F02: A	E6F03: D	E6F04: B
E6F05: A	E6F06: C	E6F07: B	E6F08: C
E6F09: D	E6F10: B	E6F11: A	